

COVID-19 Model: Scheme and Equations

1. Compartmental scheme

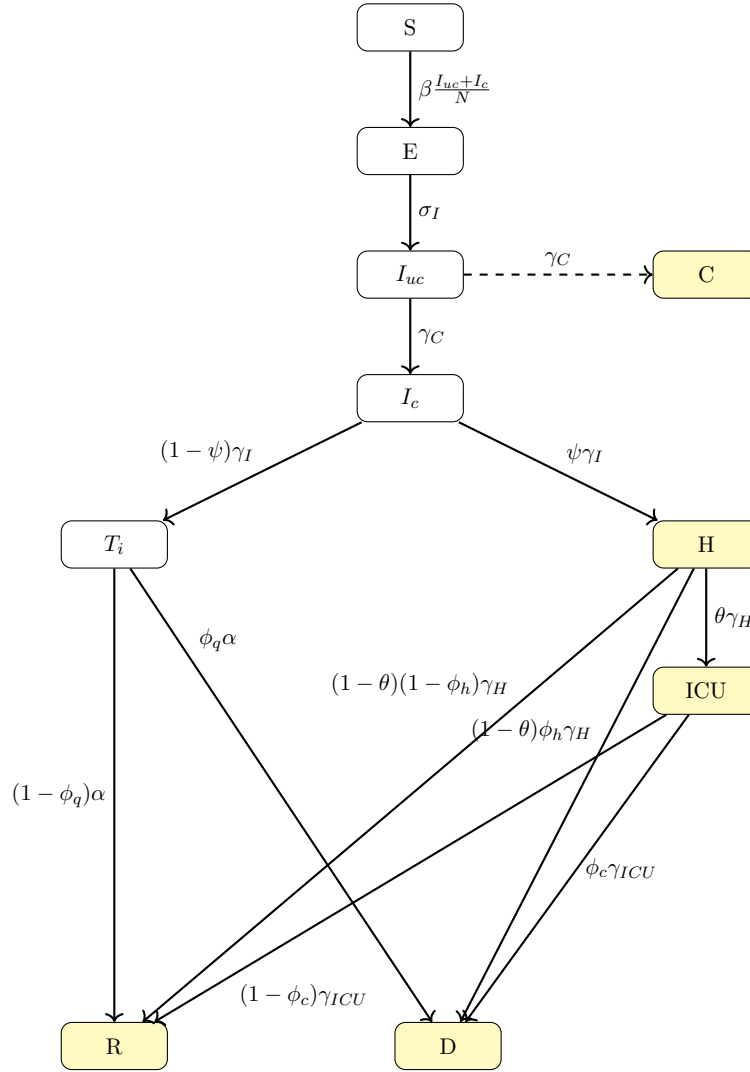


Figure 1: *

Yellow compartments are observed series: C (confirmed cases), H (ward), ICU, D (deaths). C is a non-depleting counter on the $I_{uc} \rightarrow I_c$ flow.

2. Model equations

$$\begin{aligned}
\frac{dS}{dt} &= -\beta \frac{S(I_{uc} + I_c)}{N} \\
\frac{dE}{dt} &= \beta \frac{S(I_{uc} + I_c)}{N} - \sigma_I E \\
\frac{dI_{uc}}{dt} &= \sigma_I E - \gamma_C I_{uc} \\
\frac{dI_c}{dt} &= \gamma_C I_{uc} - \gamma_I I_c \\
\frac{dT_i}{dt} &= (1 - \psi) \gamma_I I_c - \alpha T_i \\
\frac{dH}{dt} &= \psi \gamma_I I_c - \gamma_H H \\
\frac{dICU}{dt} &= \gamma_H \theta H - \gamma_{ICU} ICU \\
\frac{dR}{dt} &= (1 - \phi_q) \alpha T_i + (1 - \phi_h) \gamma_H (1 - \theta) H + (1 - \phi_c) \gamma_{ICU} ICU \\
\frac{dD}{dt} &= \phi_q \alpha T_i + \phi_h \gamma_H (1 - \theta) H + \phi_c \gamma_{ICU} ICU \\
\frac{dC}{dt} &= \gamma_C I_{uc}
\end{aligned}$$

3. Testing-volume coupling

$$Ch(t) = \exp\left(-F_{\text{test}} \cdot \frac{NT(t)}{12632}\right)$$

where $NT(t)$ is the 7-day rolling sum of daily tests, and 12632 is the mean of $NT(t)$ over 2020. $Ch(t) \in (0, 1]$ falls as testing volume grows, scaling down the hospitalisation probability below.

4. Hospitalisation probability $\psi(t)$

Model 1 (constant): $\psi(t) = \psi$

Model 2 (piecewise): $\psi(t) = \psi_k$ for t in segment k (one value per NPI-changepoint segment)

Model 3 (testing-covariate):

$$\psi(t) = \psi_0 + \psi_1 \cdot Ch(t)$$

Age-stratified model:

$$\psi(t) = Ch(t) \cdot \sum_a P_a(t) \psi_a, \quad a \in \{0-49, 50-59, 60-69, 70+\}$$

$$P_a(t) = \frac{c_a(t)}{\sum_b c_b(t)}, \quad \psi_{50-59} = 2 \psi_{0-49}, \quad \psi_{60-69} = 4 \psi_{0-49}, \quad \psi_{70+} = 8 \psi_{0-49}$$

where $c_a(t)$ is the number of age-labelled confirmed cases in group a on day t , and ψ_{0-49} is the single estimated base rate. The testing-volume coupling $Ch(t)$ above is optional on top of this (Model 3a includes it, Model 3b does not).